

Discrete Mathematics : Lecture 3 : Solve propositional satisfiable problems

SAT Algorithm, a basic NP-Complete problem.

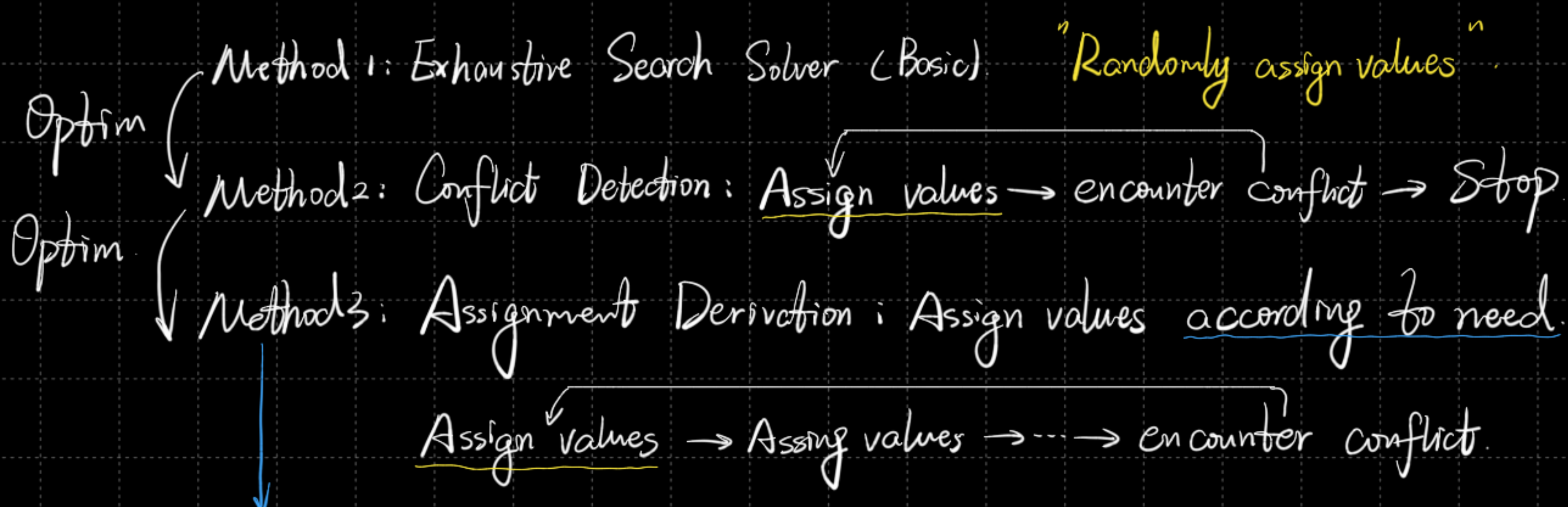
- Literal: a variable x or its negation.
- clause: A disjunction of literals.
- Conjunctive Normal Form: A conjunction of literals.
- Boolean Assignment: Assign Truth value to each Literals.

In SAT problems: Boolean variables are represented by numbers.

Subset representation:

"[,]": or " , " : and " - " : negation

Satisfiability condition: There exists an assignment where at least one literal in each clause evaluates to true.



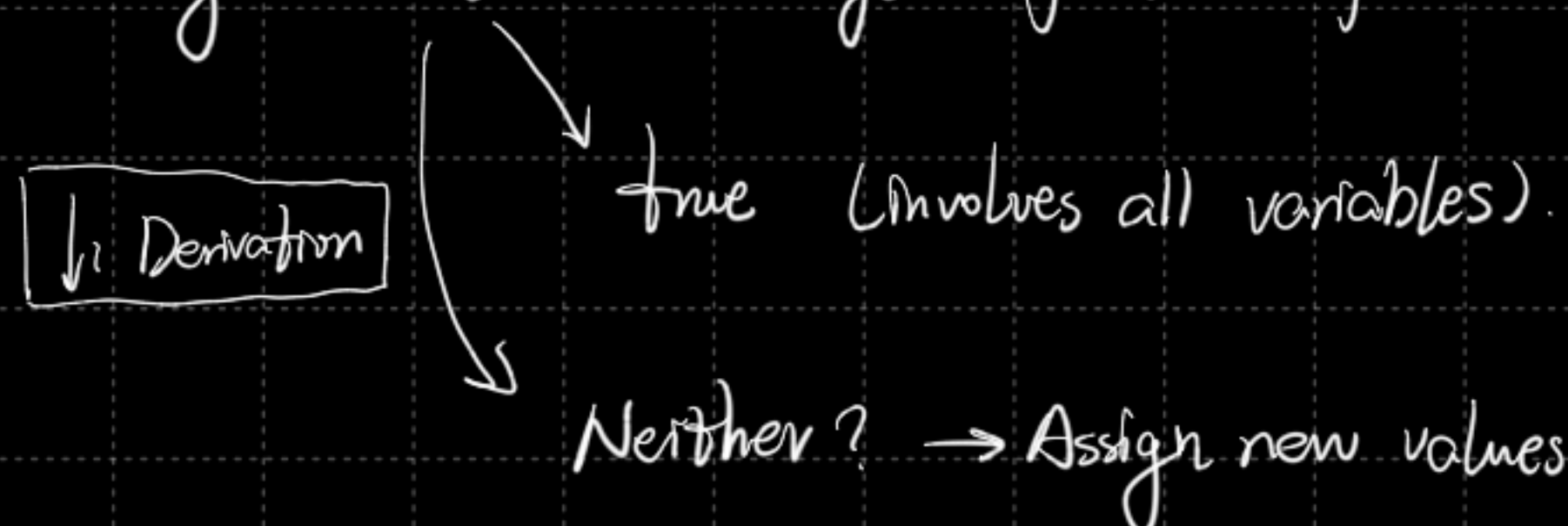
Unit Propagation: If all the other literals is false, then the last should be true

Unit Propagation: If one is true, we can ignore other literals in the clauses

Pure literal elimination: If a variable appears in all literals and clauses with a unique form,

We can assign it and ignore it.

DPLL method: Assign values → always conflict. → false



Probing: whether $x = \text{ForT}$, $y = \square$, then $\square$ might included in the potential solutions.

Variable Selection in Derivation: Choosing which to assign first matters.

- { clause based: Select the shortest clause ...
- { history based: the one cause conflicts ...

CDCL method: Conflict Driven Clause Learning

- Add SAT to "explode" conflict at first

Once we found conflicts: Add SAT to cause conflict early

To fasten the solver.

Memory: You won't fail on the same mistakes again!